

# PRAJWAL KUMAR K

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[LinkedIn](#) | [Portfolio](#) | [GitHub](#) | [Codeforces](#) | [CodeChef](#) | [LeetCode](#) | [AtCoder](#)

## EDUCATION

### Indian Institute of Information Technology, Kottayam

B.Tech in Computer Science and Engineering; CGPA: 8.66/10.0

Kottayam, Kerala

Aug 2024 – May 2028 (expected)

## TECHNICAL SKILLS

**Programming:** C++, Rust, Python, TypeScript, JavaScript, C, Java, SQL, Solidity, MATLAB

**AI/ML:** PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Matplotlib

**Research and engineering:** Jupyter, Simulink, FastAPI, Next.js, SQL

**Research interests:** Algorithms and systems, game theory, machine learning and NLP, AI safety

**Spoken Languages:** English, German, Hindi, Kannada

## EXPERIENCE AND LEADERSHIP

### Full-Stack Web Developer Intern

Dec 2025 – Mar 2026

CIO Mogul (Magazine)

Remote

- Built CIO Mogul, CIO Mogul India, and CIO Healthcare websites for publishing articles and digital issues.
- Developed Next.js/React editorial dashboards and Supabase/PostgreSQL content workflows; implemented PDF page navigation, cover extraction, and A3-spread splitting.

### Openverse and Coders Club

IIIT Kottayam

- Openverse Technical Lead (May 2025–Jul 2026): reviewed pull requests across the club and coordinated development.
- Coders Club Coordinator (Jan 2025–Present): curate problems, configure Codeforces Gym, and run campus contests.

## PROJECTS

### PokeForge | Python, PyTorch, Gen 9 OU

[Code](#)

- An agent-evaluation testbed for competitive Generation 9 OU Pokémon battles (in progress).
- Integrated pinned FoulPlay and Metamon baselines in isolated Pokémon Showdown environments; conformance and competitive strength remain under evaluation.

### Bluff | Imperfect-information games, opponent modeling

[Code](#)

- A repeated-play imperfect-information card game for studying bluff detection and adaptation to opponents.
- Built Bayesian opponent profiles and a learned public-information bluff detector; removing the detector lowered draw-aware score by 4.9–5.8 percentage points across eight synthetic profiles.

### Drishti-XAI | Retinal imaging, MATLAB, Python

- An explainable retinal-screening prototype evaluated for generalization across diabetic-retinopathy image datasets.
- Built the MATLAB/Python prototype and evaluation pipeline; at a fixed model and threshold, sensitivity/specificity were 0.908/0.877 on APTOS (n=733) and 0.363/0.865 on Messidor-2 (AUC 0.666).

### SiliconSpire | C++, Rust, Metaheuristics

[C++](#) | [Rust](#)

- A Quadratic Assignment Problem solver that minimizes weighted flow-distance cost for facility placement.
- Combined Grey Wolf Optimization with swap-based Tabu Search; evaluated C++ and Rust implementations on three small QAPLIB instances, 30 seeds each.

### BattleCP | Rust, Axum, Tokio, WebSockets

[Live](#) | [Code](#)

- A multiplayer Battleship game where solving Codeforces problems earns players turns.
- Built a Rust/Axum WebSocket backend with Codeforces validation, server-side rules, and reconnection.

## ACHIEVEMENTS AND CERTIFICATIONS

**Publication:** First-listed author and presenter, “Collaboration-Enabled Energy-Aware Learning Model Exchange (CE-LME) Framework for Edge Computing-Assisted Applications,” IEEE ICICI 2025. [Paper](#).

**Algorithmica:** selected for and presented at OOSC 3.0, IIT Kanpur (2025); team presentation at T4E 2025, IIT Madras.

**Hackathon:** Best Software Award, IEEE MACE .hack(); 2026, Team Hogorithm (Anchor).

**Competitive programming:** Codeforces Pupil (peak 1221), CodeChef peak 1463, 200+ problems solved.

**Cyfrin Updraft (2025):** Blockchain Basics, Rust Programming Basics, Solidity Smart Contract Development, and Fundamentals of Zero-Knowledge Proofs.